Lab 6: Radiometer

Measure rotation from video and compare illumination conditions. Submit the separate Lab 6 report on Canvas by Monday, December 7, 2026, 11:59 PM.

Why engineers study this

Temperature differences can drive gas motion and forces in rarefied environments. The radiometer makes this coupling visible. Video timing also develops a useful measurement skill for rotating machinery and optical tachometers. This sealed device demonstrates thermal gas forces; it is not a calibrated optical power meter.



Course apparatus from a prior test: sealed radiometer and lux meter. Green circles belong to an earlier image-analysis overlay, not physical measurement marks.

Equipment and preparation

Radiometer, existing lamp, camera and tripod, lux meter, ruler, and MATLAB. Keep the bulb sealed and upright. Do not touch a hot lamp or place it against the glass. Keep the lamp type, orientation and background fixed. Record the camera capture fps separately from the exported movie playback fps.

Before your section: complete Pre-Lab 6 and submit one Word/PDF file. No additional hardware or vacuum adjustment is required for this procedure.

1. Set up and check the video

1. Place the camera so the vanes and central axis remain visible. Focus on the vanes. Move the lamp or camera slightly to keep glare off the crossing location. Lock exposure and gain. Keep the lux meter near the rotor plane, facing the lamp, without blocking the light.

2. Record actual capture fps, lamp identity/setting, lamp-to-rotor distance, measured lux, and start/stop frames. Use enough light and a short exposure to resolve individual crossings. Check a short clip frame by frame before recording all runs.

3. Observe one complete revolution and choose one repeatable event at a fixed reference line. If a particular vane is uniquely recognizable, count that vane once per turn, $p = 1$. If counting several equivalent vane passages per turn, verify p over a full revolution. Count the same direction and appearance each time. Do not assume four visible vanes automatically mean $p = 4$.

2. Collect a small, repeatable test matrix

4. Choose three safe lamp distances that produce visibly different lux readings. Keep lamp power and type fixed. Record the actual distances and lux; do not use nominal distance as a substitute for a light measurement.

5. At each condition, record the start-up after switching on illumination and continue until the rotation is approximately steady. Start with 60 seconds and extend if needed. If memory is limited, save a start-up clip and a later steady clip and record their start times. Record lux during the steady interval.

6. Repeat each condition three times. Between repeats let the rotor and bulb return to a comparable initial condition. Record whether the rotor has stopped and how long the lamp was off. If it cannot cool comparably, report that limitation. Do not claim a settled speed from a start-up-only record.

7. Before leaving, show the GTA one film, your verified p, and at least ten consecutive crossing frames. Save raw movies and settings. Check that the crossings can actually be distinguished.

Run log

One row per run: run ID, lamp ID/setting, distance [m], lux, capture fps, playback fps, p, initial condition, illumination-on frame, selected steady frames, and filename. A CSV run-log template is included.

3. Obtain the report figures in MATLAB

8. Unzip the MATLAB bundle. Run Mark_Lab6_Passage_Frames.m. Select the original movie. Right/Left move by one step; Up/Down change step; Space marks the crossing; Backspace undoes the last mark; Q saves the passage CSV. At a crossing use a one-frame step. The first movie frame is numbered 0. The tool does not infer missing revolutions.

9. Mark consecutive crossings using the definition you checked in step 3. Run Run_Lab6_Radiometer.m, select the passage CSV, and enter actual capture fps and p. The code exports cumulative revolutions and interval rpm, plus a summary and CSV. Red circles indicate a sudden interval change that needs video review. Review it rather than forcing a smooth curve through it.

10. Repeat the analysis for a settled interval in each run. Plot its mean rpm against measured lux, showing all repeats. The mean refers to that selected interval only. The exported time starts at the first marked crossing. Use the logged illumination-on frame if discussing the physical start-up time.

Equations you need

$$t = (n - n_0)/f_c$$

n = original frame number; n_0 = reference frame; f_c = actual capture rate [frames/s]; t = elapsed physical time [s].

$$\text{rpm} = \frac{60\,\Delta N}{p\,\Delta t}$$

ΔN = number of intervals between marked events, equal to marked-event count minus one; p = verified passages per revolution; Δt = elapsed time [s]. Angular speed ω [rad/s] = 2π × rpm / 60. The result is speed magnitude. Determine rotation direction by watching the film.

4. Interpret and report

Include: (1) a labeled setup photo and settings, (2) cumulative revolutions and interval rpm for one run, (3) steady rpm versus measured lux with repeats, and (4) a short explanation of start-up and repeat differences. Use exported figures, clear units and measured values. Keep theory relevant.

Compare the trend with thermal absorption and gas-driven torque. Lux is weighted for human vision and is not irradiance [W/m²]. Do not infer optical efficiency, gas pressure, gas temperature or absolute torque from rpm alone. A first-order response is an illustrative approximation only when torque and loss parameters are nearly constant. Heating can make the start-up slower and more complex.

Resources: course lecture and The dynamic mechanism of a moving Crookes radiometer; A Horizontal Vane Radiometer: Experiment, Theory, and Simulation.